Universidad Externado de Colombia

Mixed Methods Procedures

- **Describe** the six characteristics of mixed methods research.
- **Create** a justification for using mixed methods research in a proposal or study.
- **Contrast** quantitative and qualitative data.
- **Identify**, for a core design, its intent, procedures for data collection, integration, meta-inferences, and validity.
- **Choose** a type of mixed methods design for a study and present the reasons for the choice.

Mixed Methods Procedures

What should be considered when "mixing" quantitative and qualitative data?

1. Both methods provide different types of information

Qualitative data are open data, while quantitative data are closed data.

2. Both methods have strengths and limitations

The goal is to combine the strengths to develop a stronger understanding of the research problem or questions, while overcoming the limitations of each approach.

Integration of methods and data must provide a stronger understanding of the problem or research question than either method alone.

Mixed Methods Procedures

- Have you defined mixed methods research?
- Do you justify the use of mixed methods design for your problem and question?
- Have/would you collected/collect both quantitative and qualitative data?
- Have you described your intention for collecting both forms of data?
- Have you identified a type of mixed methods design or a set of procedures to integrate your data?
- Have you provided a diagram of your design procedures?
- Have you identified how you will analyse your data for integration?
- Have/would you drawn/draw conclusions (or meta-inferences) from the integrated analysis?
- Have you discussed validity and ethics related to your research design?

Mixed Methods Procedures

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Characteristics of Mixed Methods Research

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Justification for the Use of Mixed Methods

3

Definitions

4

Process of Developing a Mixed Methods Study

Characteristics of Mixed Methods Research

6. **Frames the study with worldviews or theories:** these are the beliefs or values of the researcher (worldview) or models (theories) from the literature that frame the entire study.

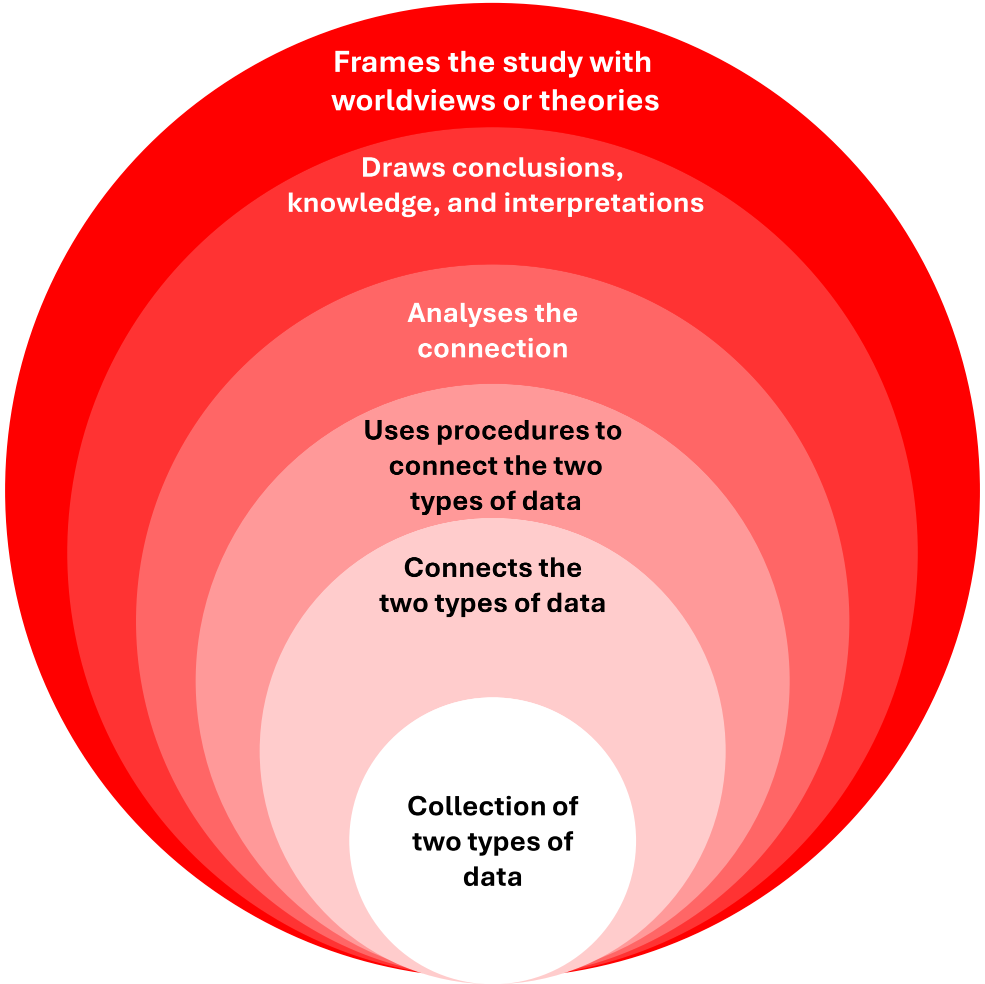
5. **Draws conclusions:** knowledge, and interpretations from the analysis, called meta-inferences.

4. **Analyses the connection** by creating a table called a joint display

3. **Uses procedures to connect the two types of data:** methods for collecting, analysing, and interpreting data, called mixed methods designs.

2. **Connects the two types of data:** "mixing" or joining the two databases, called integration.

1. **Collects two types of data:** (a) data from the participants' perspective (qualitative data) and(b) data from the researchers' or literature perspective (quantitative data)to study the problem or the research questions.



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Characteristics of Mixed Methods Research

Access to data

Researchers may have access to both quantitative and qualitative data, **which allows for additional “mining” of information beyond the results** obtained from quantitative or qualitative datasets

- A useful addition if you are trained in quantitative research is to **collect qualitative data to provide evidence that supports numerical findings**.

Problems complexity

For example, research and reporting during the pandemic, where daily statistics on new cases were observed alongside stories of people resisting vaccination.

Added value of combining data

A mixed-methods perspective provides **richer understanding** compared with reporting only quantitative or only qualitative results.

1. Develop a comprehensive **understanding of a research problem by combining** quantitative and qualitative findings.
2. **Explain quantitative results** in more detail using qualitative data.
3. **Improve measures, scales, and instruments** by incorporating participants’ perspectives who completed the instruments.
4. **Improve experiments** or trials by incorporating individuals’ perspectives.
5. **Evaluate programmes** based on the combination of quantitative and qualitative data.
6. **Develop cases** or document diverse cases for comparison

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Definitions

Data types

Distinguishing between the two types of data is more specific than simply referring to numbers versus text, or numeric data versus narratives.

- **Some data sources may contain both closed and open elements**, such as health data that include closed scores from medical tests as well as open information from patient histories.

Open data

Information collected where the researcher provides No predefined response options, such as rating scales (“strongly agree” to “strongly disagree”).

- Applied to qualitative observations and documents, the **researcher does not use a predetermined set of response options but rather observes or examines documents to identify meaning**.

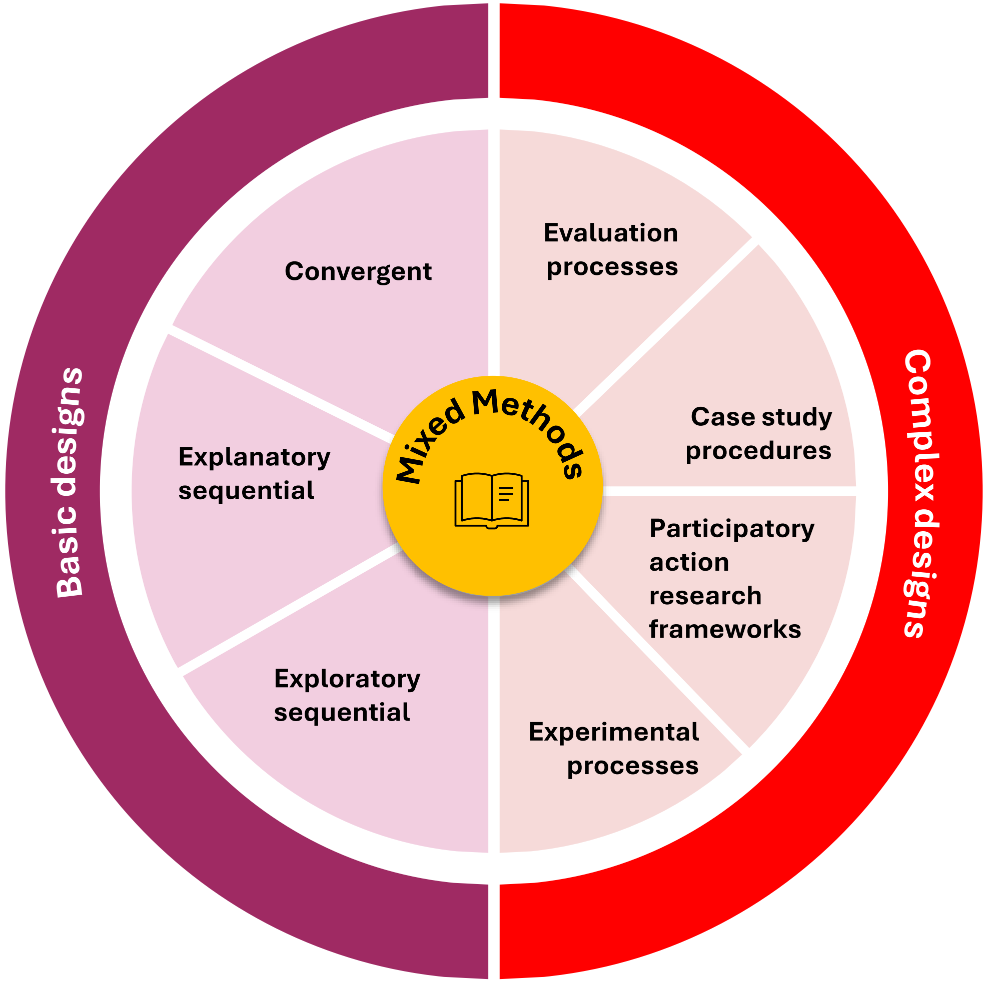
Closed data

Information collected where researchers ask a question but do not provide a response option.

Definitions

Designs

Procedure used to carry out a study, extending from broad philosophical assumptions to the interpretation of data.



Definitions

Integration

This involves combining or "mixing" quantitative and qualitative data within a study or across a series of studies.

It consists of the reasons or "intent" for combining the two databases/sources and the **procedures used to carry out this mixing**.

Integration varies depending on the type of mixed methods design. In terms of procedures:

- The researcher **merges or combines** the two databases
- The researcher **connects** them, allowing one to build upon the other.

For complex designs, integration involves **incorporating one or more basic designs into a broader process or framework**, such as an experiment, evaluation, or participatory action-research project.

Meta-inference

As the joint display table or chart is examined, **conclusions are drawn about the understanding that emerges from comparing the two databases**.

In mixed methods, these insights are called meta-inferences, meaning that a researcher reaches quantitative and qualitative conclusions and then draws additional inferences beyond what each method provides on its own.

Joint Display (visualisation)

In procedures that involve combining data, **the researcher needs some way of examining the effect of bringing the two databases together**.

A joint display is a table or chart that presents the side-by-side combination of the two databases.

- **Joint displays vary across different designs**, because the procedures for combining data differ depending on the design.

Mixed Methods Procedures

1. Characteristics of Mixed Methods Research
2. Justification for the Use of Mixed Methods
3. Definitions
4. Process of Developing a Mixed Methods Study

The process

The process of conducting a mixed methods study i**nvolves connecting quantitative and qualitative data with a specific type of mixed methods design**.

Draws a diagram of the design

Integrates the two databases

Presents the integration in a joint table or visualization for analysis

Derives insights or meta-inferences from the results of the integration

Data collection

After defining the methodology with key characteristics and a justification for its use, **it is necessary to specify the types of qualitative and quantitative data collected.**

- A useful strategy is to pose **a general problem for the mixed methods study and gather two types of data** (quantitative and qualitative) to address the issue of interest (i.e., benefits of having both open and closed information simultaneously).
- To merge the two databases during integration**, it is necessary to have a clear idea of the types of data collected**.

⚠️ You must state:

- The source of each type of data (e.g., attitude instrument, interview)
- The number of people, observations or documents collected
- Specific details about the sources (e.g., the instrument and specific scales, the online interviews).

The process for central designs

Convergent (one-phase design)

Collection of quantitative data

Collection of qualitative data

Merging of results

Interpretation of results

Explanatory sequential (two-phase design)

Collection of quantitative data

Phase 1

Identification of results

Collection of qualitative data

Phase 2

Interpretation of results

How does the qualitative e explain the quantitative?

Exploratory sequential (three-phase design)

Collection of qualitative data

Phase 1

Identification of an attribute for testing

Phase 2

Quantitative testing of the attribute

Phase 3

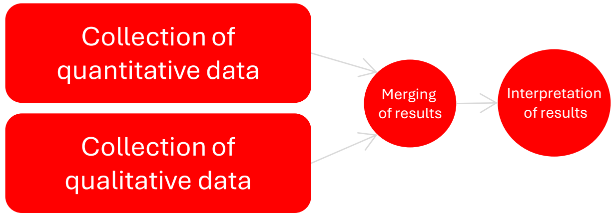
Interpretation of results

How does the testing improve our results?

Creswell (2018)

Creswell and Plano Clark identified three basic mixed methods designs: **the convergent design**, **the explanatory sequential design**, and **the exploratory sequential design**.

Convergent (one-phase design)



Key assumption:

Qualitative data and quantitative data **provide different types of information**, and their patterns/results are expected to converge.

- Detailed views from participants (qualitative)
- Scores on instruments (quantitative)

When comparing the results, the additional purpose is to assess whether the findings:

- Converge (match) or diverge
- Mutually support a construct
- Validate one form of data with the other

Campbell y Fiske (1959)

Based on the historical concept of the idea of **multiple methods and multiple traits** Campbell and Fiske (1959) suggested understanding a psychological trait by collecting different forms of data.

The goal of the convergent design is to **compare the results of the qualitative database with those of the quantitative database** by merging the results.

1. Collect quantitative and qualitative data

2. Analys the data separately

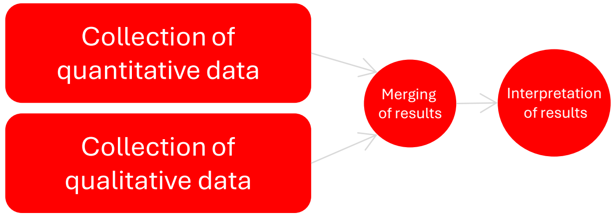
3. Compare the results to see whether the findings confirm or disconfirm each other

Ideally, both forms of data **should be collected using the same variables, constructs, or concepts**.

Measurement of the concept of self-esteem:

During quantitative data collection, questions are asked about the same concept that is also explored during the qualitative data collection process.

Convergent (one-phase design)



Sampling

Typically, mixed methods researchers include the qualitative participant sample **within the larger quantitative sample**. Ultimately, the researchers compare the two databases, and the more similar they are, the better the comparison will be.

Data analysis process

1. Analyse the qualitative database by coding the data and collapsing the codes into broad themes.
2. Analyse the quantitative database in terms of statistical results.
3. Conduct a mixed methods data analysis of the integration of the two databases. The mixed methods data analysis can be called integration analysis.

Integration

Side-by-side comparison

These comparisons appear in the discussion sections of mixed methods studies.

1. Report the quantitative statistical results.
2. Discuss the qualitative findings (e.g., themes) that confirm or disconfirm the statistical results.

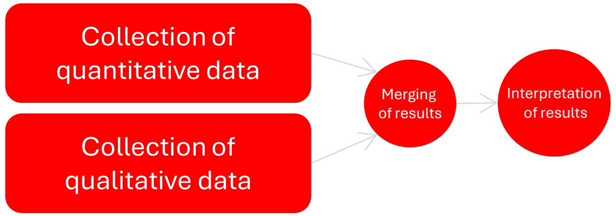
Themes-to-variables / data transformation

Merge the two databases by changing **or transforming qualitative codes or themes into quantitative variables** and then combine the two quantitative databases.

Graphical representation

Merge the two forms of data into a table or graphic. **This table or graphic is called a joint display**.

Convergent (one-phase design)



Graphical representation – Joint display

Quantitative scores	Theme 1	Theme 2	Theme	Theme 4	Meta inferences
High	Cite/Punctuations	Cite/Punctuations	Cite/Punctuations	Cite/Punctuations	Insight
Median	Cite/Punctuations	Cite/Punctuations	Cite/Punctuations	Cite/Punctuations	Insight
Low	Cite/Punctuations	Cite/Punctuations	Cite/Punctuations	Cite/Punctuations	Insight
Meta inferences	Insight	Insight	Insight	Insight	Insight

1. **Add a column and a row for meta-inferences to the themes and scores.** This allows results to be obtained by examining rows or columns containing qualitative and quantitative data.
   - How do the scores of individuals with high, medium, and low scores differ for Theme 1?
   - How do individuals with high scores differ across the four themes?

2. **Examine both databases and draw conclusions or insights from the analysis.**

3. **Look for types of integrative results,** such as whether the qualitative and quantitative results:
   - Provide confirmation (agreement)
   - Show discordance (disagreement).

4. **Relate these findings to existing literature** and/or explain how the meta-inferences inform theories.

Convergent (one-phase design)

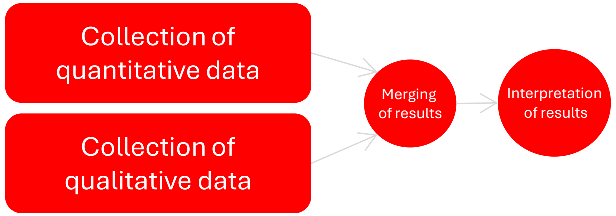


Figure 3. A cross-case comparison joint display from a convergent design showing scored items and descriptions.

Table 3. Cross-Case Comparison Using Three Participants and Mixed Methods Integration of Quantitative Scores and Qualitative Assessments

Domain	Variables	Participant 1	Participant 2	Participant 3
Knowledge cognition		DHFKS total = 14 DSST = 25, PMR = 2	DHFKS total = 14 DSST = 23, PMR = 0	DHFKS total = 15 DSST = 46, PMR = 4
Self-care maintenance	SCHFI score Diet, monitoring exercise, and medication	100 Follows low-fat and low-salt diet, fluid 2 L restriction, weighs self daily, exercises 2-3 times per week, pillboxes for medication	90 Follows 2 grams low-salt diet; takes lunch to work. Checks and records blood pressure and writes weights on calendar. Exercises on treadmill each day. Medication log	60 Low-salt diet "used to be better," now has dietary indescretions. Tries to exercise regularly but not consistent. Medication routine: Medicines make the participant tired, so sometimes "is lazy to take"
		Cheats (on diet) and manages (symptoms)	*Self-care maintenance as routine*	*Inconsistent self-care*
Self-care management	SCHFI score Symptom monitoring, symptom recognition, symptom importance, action, symptom improvement	87.57 Checks ankles and daily weights, records data, and in presence of symptoms eats less salt; diuretic titration; energy conservation. Recognizes that increased urination and weight loss indicate improvement	74.21 Daily weights, checks blood pressure symptoms such as hyperventilating. with symptoms, rests or stops activity, calls health care provider immediately. Improvement noted as breathing eases. Also has external defibrillator	67 Daily weights (or 3 times/week). "Knows body" and relies on intuition to identify symptoms. Often will just work through symptoms and wait to see if feels better. Does not pay attention to some symptoms (e.g., what is fatigue from HF, from work, and from motherhood)
		Consistent self-care Symptom vigilance	*Consistent self-care Symptom vigilance*	*Lacks vigilance Watches and waits on symtoms*
Knowledge	HF	Heart is weakened	Heart arrhythmia	Describes postpartum

Guetterman et al., (2015)

Dickson et al. (2011) **investigated how cognitive function and knowledge affected self-care in heart failure, using a convergent design.**

In two specialised outpatient heart-failure clinics in a large urban medical center in the United States, **they administered standardized instruments to measure self-care, knowledge, and cognitive function**

They simultaneously conducted interviews with patients to understand their self-care practices.

The integration focused on the **concordance between the qualitative and quantitative results**

Dickson et al. **developed a joint display comparing cases to compare interview data with quantitative scales of self-care, cognitive function, and knowledge**.

- The visualisation contained one row for each **qualitative domain** (theme) and reported the **quantitative variables** corresponding to that domain.
- The participants were arranged in columns.
- Inside each cell were the actual quantitative scores as well as summaries and qualitative quotations for each theme–participant combination.

In this example, the authors used the visualisation to validate the quantitative scores for knowledge and self-care, while also identifying cases of inconsistencies.

The research in central designs

When divergence occurs, the researcher needs to conduct a follow-up analysis.

- **Divergence may be declared as a limitation in the study** without additional follow-up. However, this approach represents a weak solution.
- The researcher may return to the analyses and **explore the databases more deeply**.
- **The researcher may gather additional information to resolve the differences or discuss limitations in one of the databases** (e.g., invalid quantitative constructs or a poor correspondence between the open-ended question and the qualitative themes).

What to do when divergence is identified?

Explanatory Sequential (two-phase design)

Collection of quantitative data → Identification of results → Collection of qualitative data → Interpretation of results — How does the qualitative e explain the quantitative?

Phase 1 | Phase 2

Key assumption:

Quantitative results may produce unexpected, unusual, significant, or atypical findings, or demographic patterns that **require additional explanation**.

- The qualitative follow-up phase provides this explanation.
- The qualitative data collection is directly based on the quantitative findings.

Sampling

Procedures must occur over time:

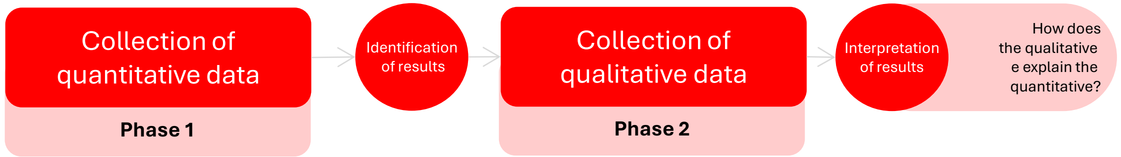
1. Quantitative sampling (Phase 1)
2. Purposive qualitative sampling (Phase 2)

The qualitative sample should be a subset of the quantitative sample. Researchers must anticipate the qualitative phase early, even before fully completing the quantitative phase.

Data analysis process

1. Collection of qualitative data to explain the quantitative results in greater detail
2. The quantitative results typically inform:
   - Which participants are purposively selected for the qualitative phase
   - Which questions are asked in the qualitative interviews

Explanatory Sequential (two-phase design)



Graphical representation – Joint display

Quantitative scores	Qualitative follow-up themes	Meta-inferences
High	Theme 1, Theme 2, Theme 3	How the themes explain the scores
Median	Theme 3, Theme 4	How the themes explain the scores
Low	Theme 5	How the themes explain the scores

In an explanatory sequential design, **the qualitative sample is a subset of the quantitative sample, leading to overlapping samples**; the integration is achieved by connecting Phase 1 results to guide Phase 2 data collection, rather than by a post-hoc side-by-side comparison.

1. **Separate analysis of quantitative and qualitative data.**

2. **Connect the databases when the quantitative results guide the qualitative data collection (integration point).** This can be done with a joint display. The template for this design shows:

- Columns for the quantitative scores
- Columns for the qualitative follow-up themes, which are based directly on the quantitative results.

The joint display, read left to right, follows the order of procedures in the design.

3. **Identify the meta-inferences** rather than confirmation or agreement**, the conclusions represent an extension of the quantitative results**.

- The results can help design future quantitative evaluations.
- The findings must be compared with existing literature and theories

Convergent (one-phase design)

Collection of quantitative data
Phase 1
→ Identification of results →
Collection of qualitative data
Phase 2
→ Interpretation of results
How does the qualitative e explain the quantitative?

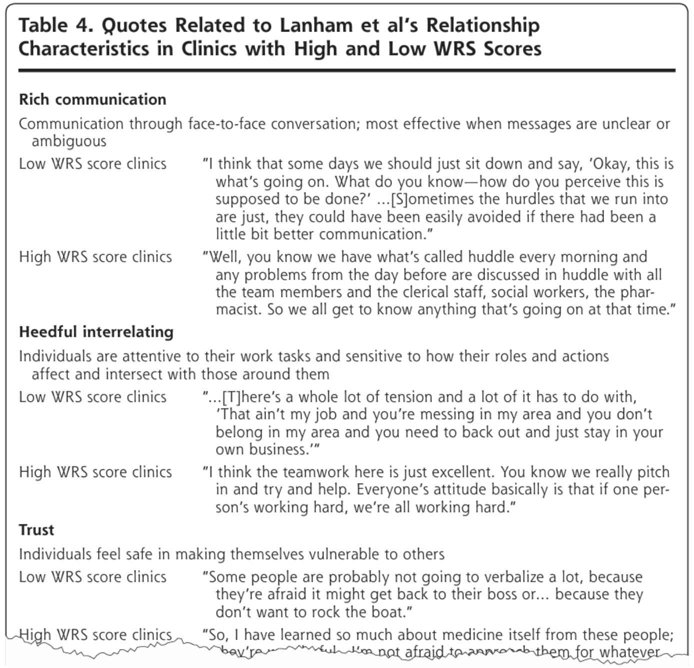


Table 4. Quotes Related to Lanham et al's Relationship Characteristics in Clinics with High and Low WRS Scores

Rich communication	
Communication through face-to-face conversation; most effective when messages are unclear or ambiguous	
Low WRS score clinics	"I think that some days we should just sit down and say, 'Okay, this is what's going on. What do you know—how do you perceive this is supposed to be done?' ...[S]ometimes the hurdles that we run into are just, they could have been easily avoided if there had been a little bit better communication."
High WRS score clinics	"Well, you know we have what's called huddle every morning and any problems from the day before are discussed in huddle with all the team members and the clerical staff, social workers, the pharmacist. So we all get to know anything that's going on at that time."
Heedful interrelating	
Individuals are attentive to their work tasks and sensitive to how their roles and actions affect and intersect with those around them	
Low WRS score clinics	"...[T]here's a whole lot of tension and a lot of it has to do with, 'That ain't my job and you're messing in my area and you don't belong in my area and you need to back out and just stay in your own business.'"
High WRS score clinics	"I think the teamwork here is just excellent. You know we really pitch in and try and help. Everyone's attitude basically is that if one person's working hard, we're all working hard."
Trust	
Individuals feel safe in making themselves vulnerable to others	
Low WRS score clinics	"Some people are probably not going to verbalize a lot, because they're afraid it might get back to their boss or... because they don't want to rock the boat."
High WRS score clinics	"So, I have learned so much about medicine itself from these people;

Guetterman et al., (2015)

Finley et al. (2013) carried out an **explanatory sequential study** with the aim of validating the Work Relationships Scale (WRS) for primary-care clinics. They analysed the measurement properties of the WRS and correlations with patients' ratings of quality of care

They interviewed key informants about relationships in the clinics and analysed the interviews based on the seven characteristics of the WRS.

They concluded **that the interview data supported the statistical analysis**, providing evidence of validity for the WRS and indicating the importance of relationships in delivering primary care.

They developed a **joint display of statistics by themes** that compared clinics with low and high WRS scores

- The visualisation had a row for clinics with low WRS with a representative quotation, and then a row for clinics with high WRS with a representative quotation.
- The headings organised the results according to the theoretical model of work relationships.

The research process in central designs

As in all mixed methods studies, the researcher needs to establish the validity of the quantitative measure scores and the qualitative findings.

- **The precision of the overall findings can be compromised** because the researcher does not consider and weigh all the options for following up on the quantitative results.
- Attention may focus only on personal demographics, **overlooking important explanations that require deeper understanding**.
- The researcher may also contribute to invalid results by **relying on different samples in each phase of the study**.
    - If the goal is to explain the quantitative results in greater depth, it makes sense to select the qualitative sample from among the individuals who took part in the quantitative sample.

Exploratory Sequential (three-phase design)

Collection of qualitative data → Identification of an attribute for testing → **Quantitative testing of the attribute** → Interpretation of results

Phase 1 | **Phase 2** | **Phase 3**

How does the testing improve our results?

Key assumption:

We assume that qualitative data can improve (and make contextualised) quantitative measures, scales, or instruments.

- Develop a survey instrument or questionnaire because none exists in the literature.
- Develop new variables not present in the literature.
- Modify the types of scales that may exist in current instruments.
- Form categories of information explored more deeply in the quantitative phase.

Sampling

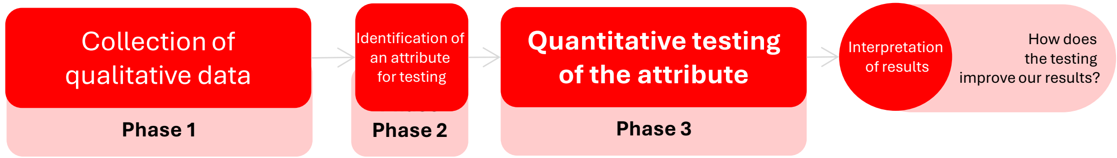
Should individuals in the qualitative sample also be individuals in the quantitative sample?

No. The qualitative sample is typically much smaller than the quantitative sample needed to generalise to a population. However, a good procedure is to draw both samples from the same population but **ensure that the individuals in each sample are not the same.**

Data analysis process

1. Collect qualitative data
2. **Connect the qualitative findings with a phase of designing quantitative instruments** (context-sensitive), or variables not available in the literature (and adapt them to a specific context).
   - **Quotations** → to write survey items
   - **Codes** → to develop variables grouping those items
   - **Themes** → to create scales grouping the codes
3. **Test the instrument using sampling/quantitative designs.**

Exploratory Sequential (three-phase design)



Graphical representation – Joint display

Qualitative insights	Qualitative follow-up themes	Meta-inferences
Quotations	Items of the quantitative survey	Analyze survey scores/indicators
Codes	Variables of the quantitative survey	Identify variables
Themes	Scales of the quantitative survey	Identify scales

1. **Separate analysis of the quantitative and qualitative data**, using the findings from the initial exploratory qualitative database to construct a characteristic for the quantitative analysis.

   Integration in this design involves using qualitative findings (or results) to inform the design of a quantitative phase of the research.

2. **Design of attributes / instruments**

3. **Implementation of the quantitative stage**

4. **Development of meta-inferences.** examine the qualitative characteristics and the design characteristics to test the adapted quantitative evaluation.
   - How well has the adaptation occurred?
   - Is the modified or newly designed survey producing good results?
   - Will the test show sensitivity to the sample and population under study?

Exploratory Sequential (three-phase design)

Collection of qualitative data
Phase 1

Identification of an attribute for testing
Phase 2

Quantitative testing of the attribute
Phase 3

Interpretation of results
How does the testing improve our results?

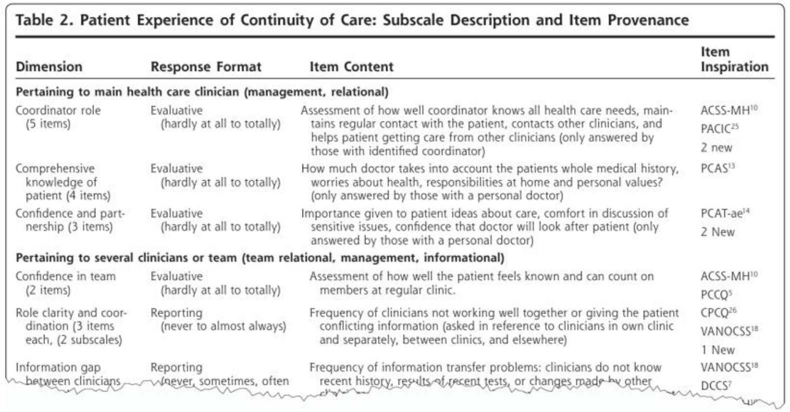


Table 2. Patient Experience of Continuity of Care: Subscale Description and Item Provenance

Dimension	Response Format	Item Content	Item Inspiration
Pertaining to main health care clinician (management, relational)			
Coordinator role (5 items)	Evaluative (hardly at all to totally)	Assessment of how well coordinator knows all health care needs, maintains regular contact with the patient, contacts other clinicians, and helps patient getting care from other clinicians (only answered by those with identified coordinator)	ACSS-MH[10] PACIC[25] 2 new
Comprehensive knowledge of patient (4 items)	Evaluative (hardly at all to totally)	How much doctor takes into account the patients whole medical history, worries about health, responsibilities at home and personal values? (only answered by those with a personal doctor)	PCAS[13]
Confidence and partnership (3 items)	Evaluative (hardly at all to totally)	Importance given to patient ideas about care, comfort in discussion of sensitive issues, confidence that doctor will look after patient (only answered by those with a personal doctor)	PCAT-ae[14] 2 New
Pertaining to several clinicians or team (team relational, management, informational)			
Confidence in team (2 items)	Evaluative (hardly at all to totally)	Assessment of how well the patient feels known and can count on members at regular clinic.	ACSS-MH[10] PCCQ[5]
Role clarity and coordination (3 items each, (2 subscales)	Reporting (never to almost always)	Frequency of clinicians not working well together or giving the patient conflicting information (asked in reference to clinicians in own clinic and separately, between clinics, and elsewhere)	CPCQ[26] VANOCSS[18] 1 New
Information gap between clinicians	Reporting (never, sometimes, often	Frequency of information transfer problems: clinicians do not know recent history, results of recent tests, or changes made by other	VANOCSS[18] DCCS[7]

Guetterman et al., (2015)

Haggerty et al (2012) carried out a sequential exploratory design to **develop and validate an instrument that evaluates the continuity of care from the patients' perspective**. They examined themes from 33 qualitative studies on patient-care experiences.

A joint display for instrument development **mapped the qualitative dimensions of continuity of care to the items of the quantitative instrument**.

- The main row headings marked each dimension of continuity of care.
- The columns provided the response format, a description of the item content, and the source of the item (i.e., existing survey or new item).
- By presenting each qualitatively derived dimension together with the corresponding instrument item, the **display clearly showed how the instrument was systematically developed**.

The research process in central designs

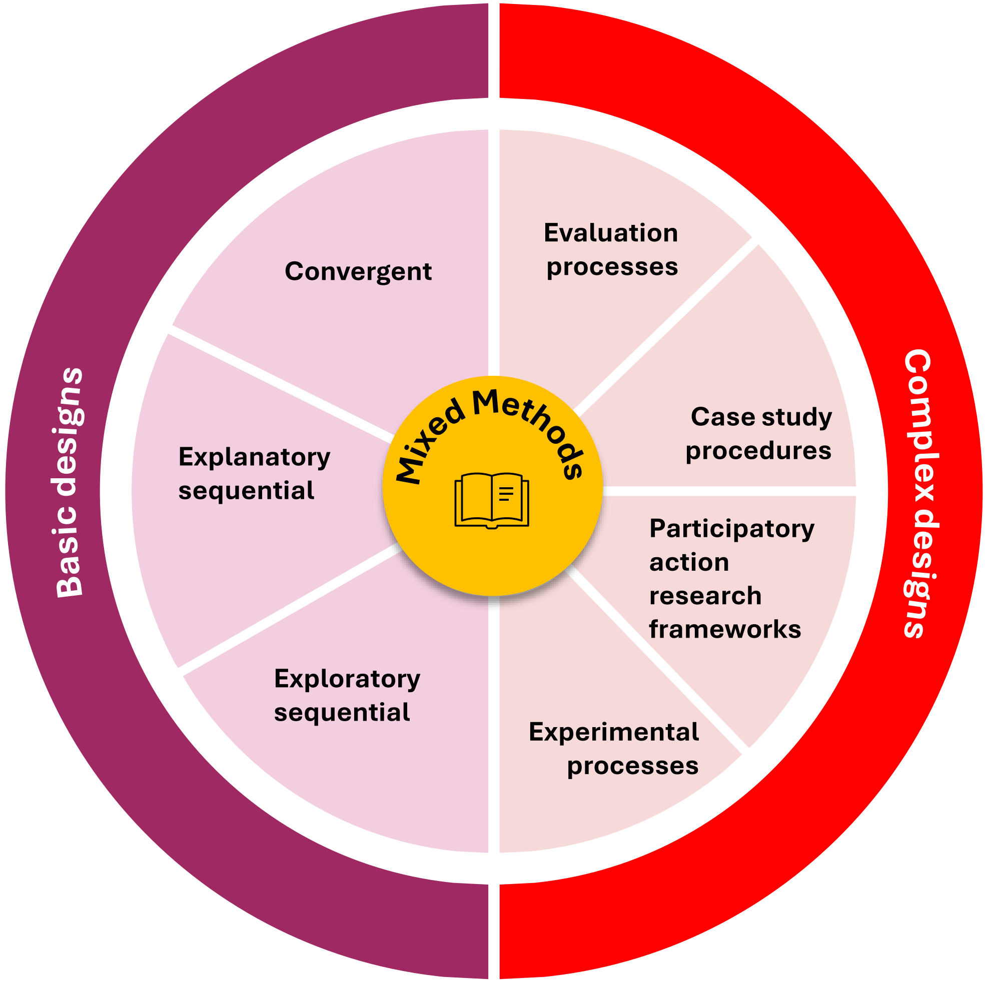
As in all mixed methods studies, the researcher needs to establish the validity of the quantitative measure scores and the qualitative findings.

- **Failure to use the appropriate steps to develop a good psychometric instrument.**
- Development of an instrument or measures **that do not make full use of the richness of the qualitative findings**.
- The sample in the qualitative **phase should not be included in the quantitative phase**, because this would introduce undue duplication of responses.
- Therefore, this sampling strategy differs from the sampling strategy needed for an explanatory sequential design.

Definitions

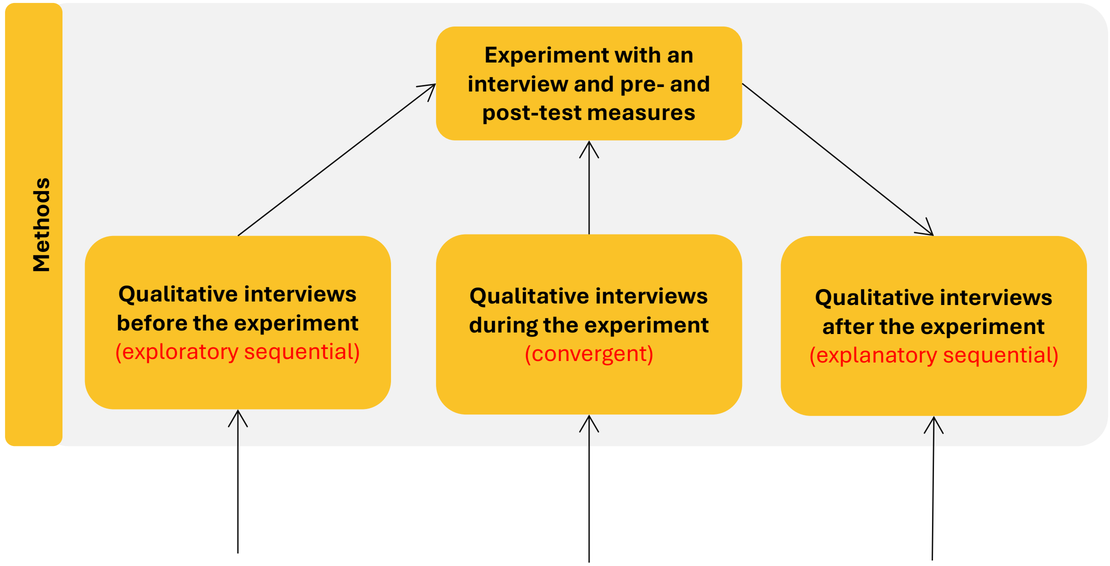
Designs

Procedure used to carry out a study, extending from broad philosophical assumptions to the interpretation of data.



The research process in complex designs

Mixed Methods Experimental Design



- Recruitment of participants
- Development of applicable interventions
- Development of pre-test and post-test measures

- Examination of participants’ experiences
- Modifications to the treatment

- Examination of the results
- Modifications to the experiment

Integration of qualitative and quantitative data

This design allows the researcher to collect and **analyse both qualitative and quantitative data within the same study**.

- Qualitative data are integrated at different points of the experiment to capture participants’ personal experiences.
- This enriches the analysis and provides a more complete understanding of the phenomenon.

Support for experimental data collection

Qualitative data support the pre-test and post-test measurements of the experiment

- By adding qualitative data, researchers can obtain a **deeper understanding of how and why participants respond in certain ways to the quantitative tests**, helping interpret the results more effectively.

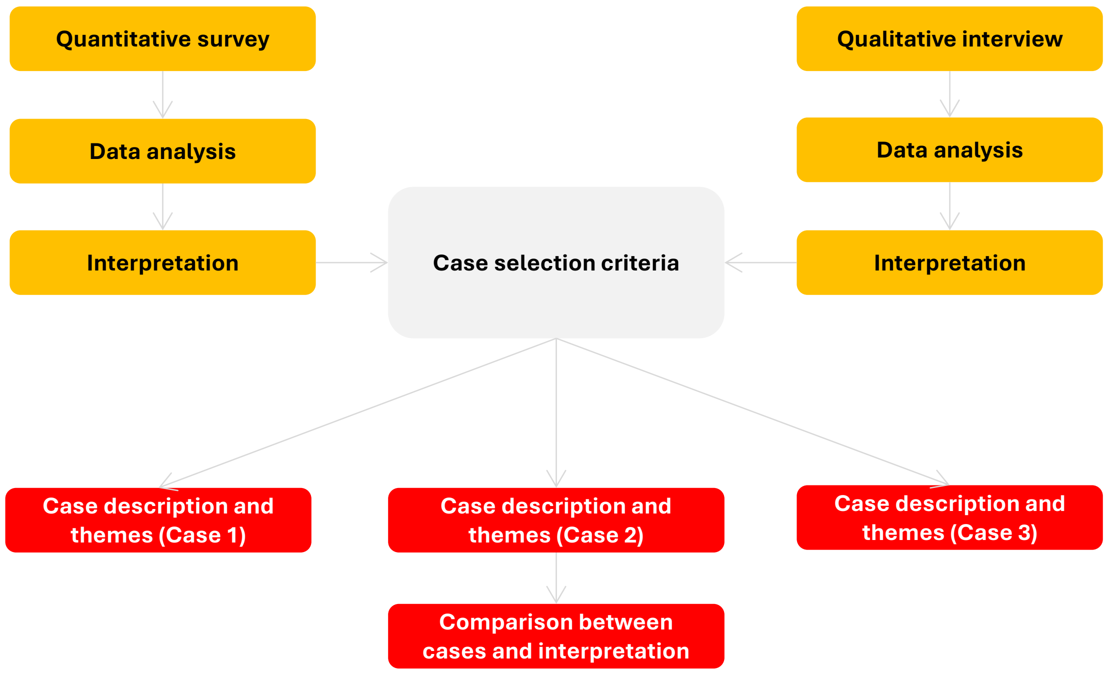
Different forms of integration

Qualitative data may be added before, during or after the experiment.

- **Before** → qualitative data help design the intervention and understand participants’ initial context.
- **During** → qualitative data provide real-time information about implementation and participant experience.
- **After** → qualitative data help interpret results and explore long-term impact and sustainability.

The research process in complex designs

Mixed Methods Case Study Design



In this design, **the researcher integrates a central design within the broader process of developing case studies for analysis**.

- The central design is a convergent design
- The process consists of developing cases inductively for their description and comparison.

Deductive approach

- **The cases are established at the beginning of the study.**
- **Differences between the cases are documented through qualitative and quantitative data.**

Inductive approach

- **The researcher collects and analyses both types of data.**
- **Cases are formed during the research process based on the evidence gathered.**

Description and Comparison of Cases

- **Describing each case**
- **Then discussing the similarities and differences between cases**

Challenges in Case Identification and Understanding

- **Identifying the cases before the study begins**
- **Generating cases based on the evidence collected**

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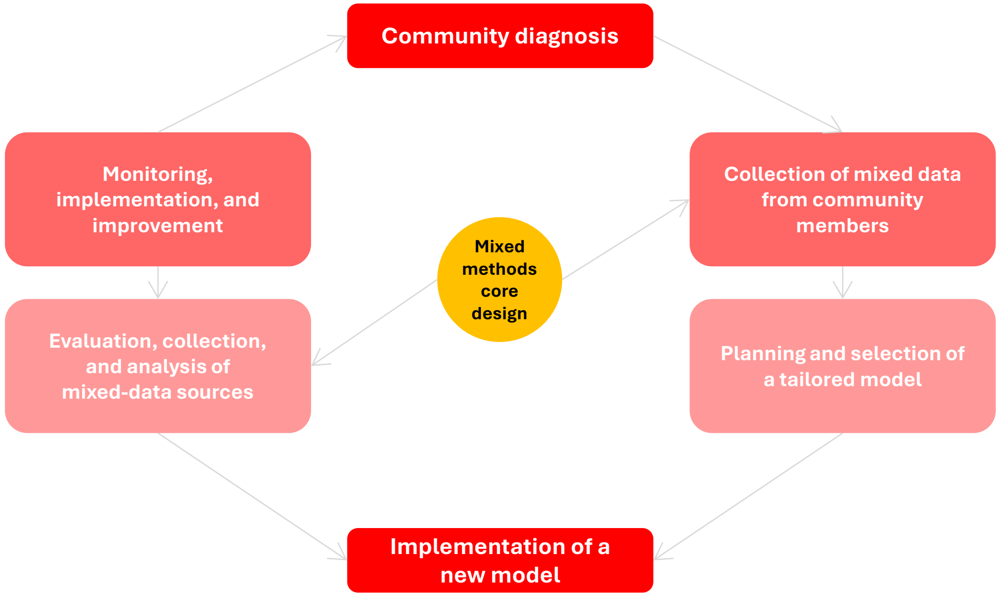
Participatory Action Research Design in Mixed Methods



Active community participation

Community members actively participate in the research. **They collaborate closely with researchers** throughout the entire research process.

Focus on social change

Social-justice studies aim not only to gather data but also **to promote significant social change and actions that improve people's lives**. This approach goes beyond data collection and focuses on meaningful, concrete actions for the community.

Collaborative evaluation process

- An evaluation to diagnose the community's needs
- Data collection through community assessments
- Identification and implementation of a model that meets those needs
- Continuous evaluation of the model's success

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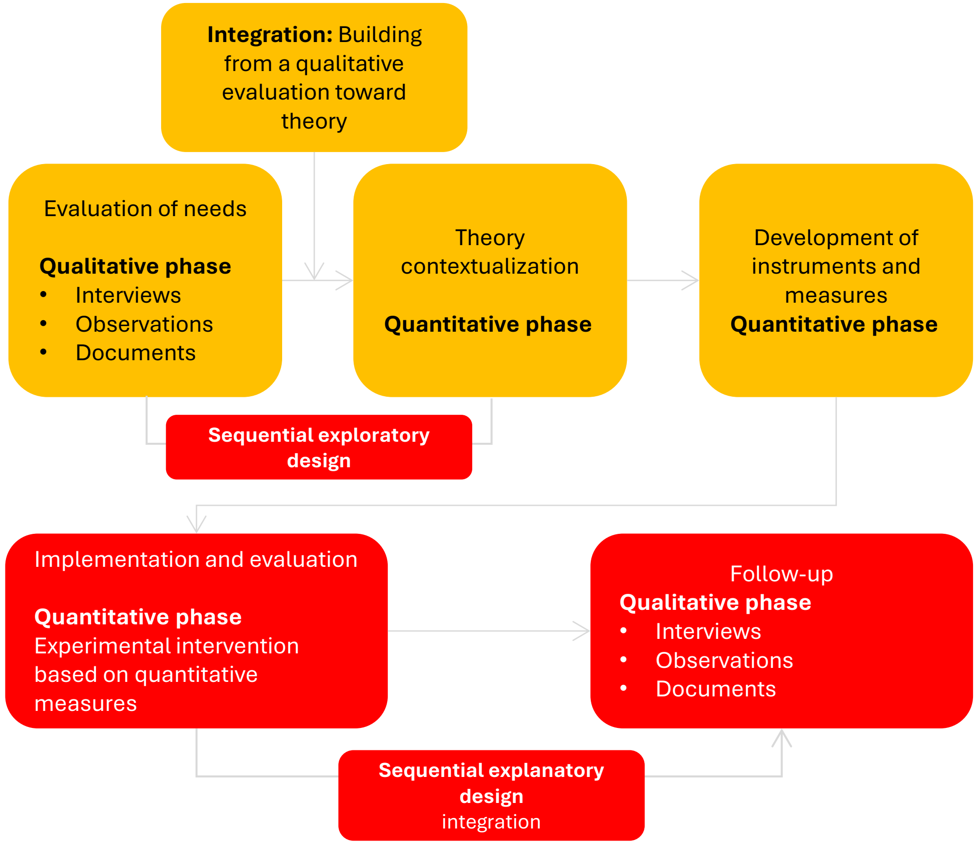
Mixed Methods Evaluation Design



Integration of quantitative and qualitative data

This design combines quantitative and qualitative data within an **evaluation framework** to provide a more complete and detailed understanding of the programme, experiment, or policy being evaluated.

Central designs

Across these projects, multiple central designs are often used.

- When evaluating a programme, it is possible to begin by collecting qualitative data in a needs assessment.
- Based on this assessment, a conceptual model is specified and instruments are developed.
- The programme is then implemented and tested, followed by a qualitative follow-up phase to refine the programme.

Combining exploratory and explanatory approaches

It is possible to use multiple central designs.

- Moving from a needs assessment to theoretical conceptualisation requires combining qualitative data with a quantitative evaluation in a sequential exploratory design.
- After implementing and quantitatively testing the programme, a qualitative follow-up phase (or the integration of a sequential explanatory design) is added.

The research process in central designs

The intention for using a design and the procedures for carrying out the research differ depending on whether the **design is central or complex**.

The aim of this design is to carry out a research process in which both **quantitative and qualitative data and their integration shape one or more stages of the process**.

How to develop a justification for the choice of design type?

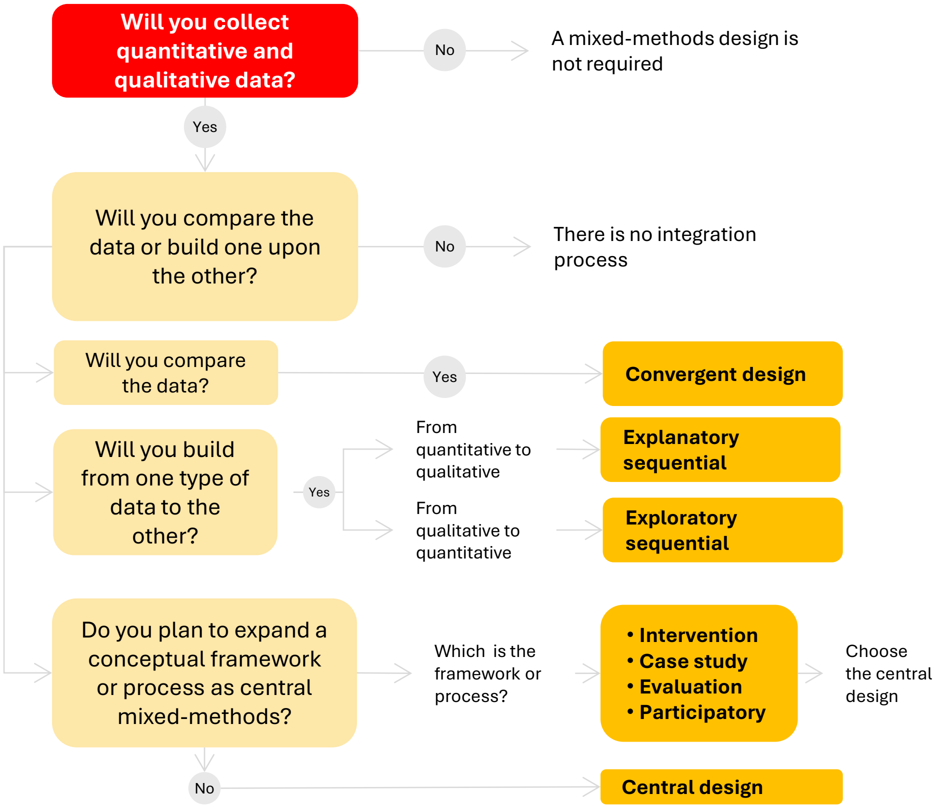
The intention for using a design and the procedures for carrying out the research differ depending on whether the design is central or complex.

Design	Intention / Purpose	Procedure
Convergent	Compare, match, corroborate (validate), expand, enhance, diffract, identify cases, initiate, full understanding	**Connect** (qualitative results lead to the design of a quantitative evaluation that is then tested)
Explanatory sequential	Expand, explain	**Connect** (qualitative results lead to the design of a quantitative evaluation that is then tested)
Exploratory sequential	Construct, transfer, generalise	**Connect** (qualitative results lead to the design of a quantitative evaluation that is then tested)
Complex	Expand (optimise, determine needs, monitor)	**Integrate** qualitative, quantitative, or both data types into a process or framework

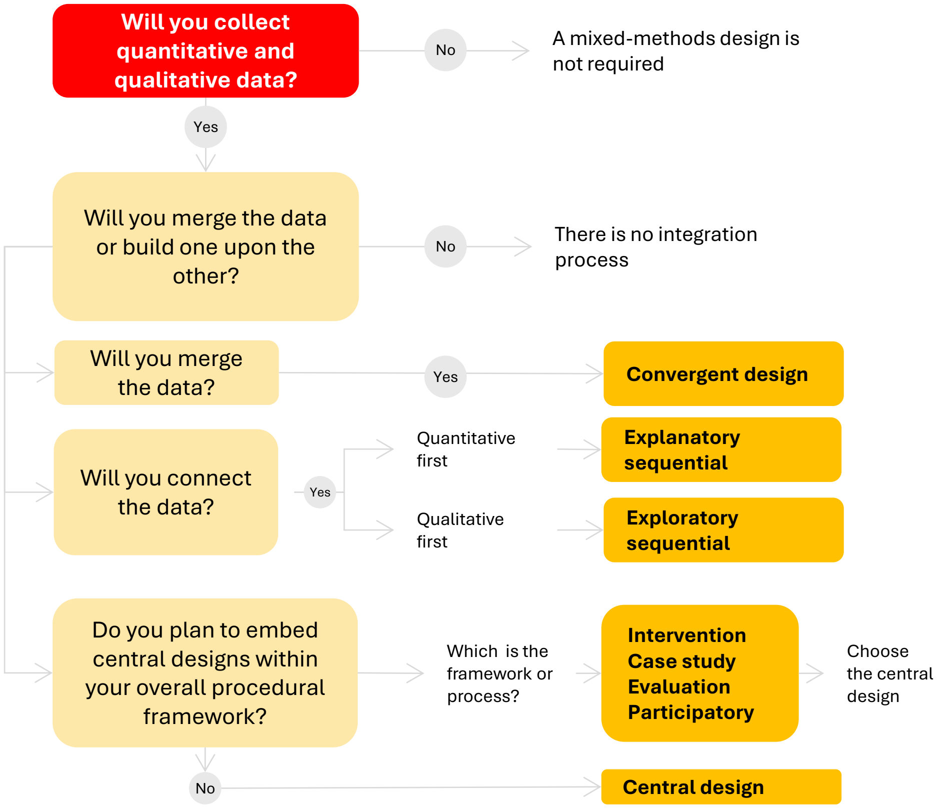
- **Convergent design:** Integration involves comparing the results of the quantitative and qualitative data by merging them, so that a more complete understanding emerges than that provided by either quantitative or qualitative results alone.
- **Explanatory sequential design:** Integration involves explaining the results of the initial quantitative phase by connecting it to a qualitative phase. This connection includes which questions need further exploration and who can best help explain the quantitative results.
- **Exploratory sequential design:** Integration involves initial exploration by collecting qualitative data, analysing it, and using the qualitative results to build a culturally specific measure or instrument for quantitative testing with a large sample.

Selection of designs based on purposes and procedures

Flowchart for choosing your type of design (based on purposes)



Flowchart for choosing your type of design (based on procedures)



Selection of designs based on purposes and procedures

1. Identify the collection of quantitative and qualitative data in your study.
2. Indicate whether the data source is quantitative or qualitative.
3. Draw a diagram of the steps in the procedure. These steps (represented by boxes) show the phases of the study.
4. Look at the steps (boxes) and ask yourself in which steps of the procedure you could collect both quantitative and qualitative data. This data collection represents a central feature of mixed-methods research.
5. In the boxes where you are collecting both types of data, also ask yourself how those data are being connected.
    - Merged (as in a convergent mixed-methods design)
    - Connected (as in an explanatory sequential or exploratory sequential mixed-methods design)
6. Discuss the procedures of the basic mixed-methods designs and how they would apply, paying attention to how the data are being integrated at each step.